ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION 2024-25

B. Tech. (CSE)/ B.Tech. & M.Tech. (CSE)/ B.Tech. & MBA

Semester: VI

SUBJECT CODE: BCS602

COMPILER DESIGN

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Answer the following questions: **(1 × 5= 5 Marks)**

(i) List out the rules for FIRST and FOLLOW.
(ii) Compare synthesis and inherited translation.
(iii) Draw the diagram for the language processing system.
(iv) Explain the advantage of inter procedural optimization.
(v) What are the actions performed by Shift reduce parser?

Q. 2. Answer the following questions: **(1 × 5= 5 Marks)**

(i) Define Boot strapping.
(ii) Give the structure of symbol table.
(iii) What are the draw backs of predictive parsing?
(iv) What is the role of type system in type checker?
(v) How the Garbage collection works through reference counting?

SECTION – B

NOTE:Attempt all questions. **(5×2×2 = 20 Marks)**

Q.3.(i)(a) Explain the phases of the complier in detail. Write down the output of each phase for the expression **a=b*c+50.** Standard precedence for operation may be used.

OR

(b) What are different ways to write three address codes? Write the three-address code for the following code segment:

While (a<c AND b<d) do
If a=1 the c=c+1
Else while a<=d do a=a+2

(ii)(a) Consider the following grammar and give the syntax directed definitions to construct parse tree. For the input expression **4*7+1*2,** construct an annotated parse tree according to your syntax directed definition:

E→ E+T/T
T→ T*F/F
F → Digit

OR

(b) Explain different types of intermediate code forms and represent the following statement in different forms:

W = (A + B) – (C + D) + (A + B + C).

Q.4.(i)(a) What is DAG? How DAG is created from three address code? Write algorithm for it and explain it with a relevant example.

OR

(b) What is meant by activation of procedure? How it can be represented with activation tree and record?

(ii)(a) Explain three techniques for loop optimization with examples.

OR

(b) What is the role of symbol table? Discuss different data structures used for symbol table.

SECTION – C

NOTE: Attempt all questions. **(10 × 2 = 20Marks)**

Q.5. (a) Construct CLR Parsing table for the grammar and check the acceptance of input string **id=id/id.**

S→L=R/R
L→*R/id
R→L

OR

(b) What are the preprocessing steps required for predictive parse table construction? Consider the grammar:

S→ACB|CbB|Ba
A→da|BC
B→g|ε
C→h| ε

Q.6. (a) Construct an SLR (1) parsing table for the following grammar:

S→A)
A→ A, P| (P, P
P→ {num, num}

OR

(b) Construct the Deterministic finite automata for the regular expression: **10+ (0+11)0*1**